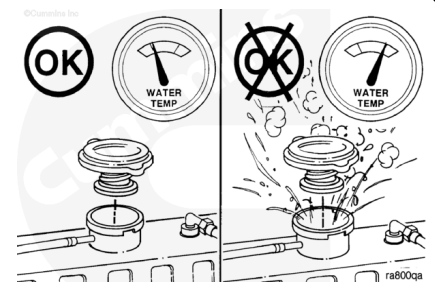


Trainings ▾

Drain

⚠ WARNING ⚠

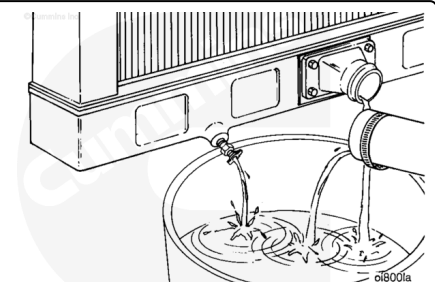
Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



Remove the radiator cap after the engine is cool.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.



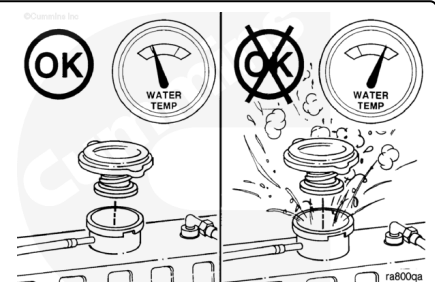
Position the vehicle on level ground. Open the draincock at the bottom of the radiator. Remove the lower radiator hose. Drain the cooling system.

Do **not** remove the coolant filter.

Flush

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



The performance of RESTORE™ is dependent on time, temperature, and concentration levels. An extremely scaled or flow restricted system, for example, can require higher concentrations of cleaners, higher temperatures, or longer cleaning time. RESTORE™ can be safely used up to twice the recommended concentration levels. Extremely scaled or fouled systems can require more than one cleaning.

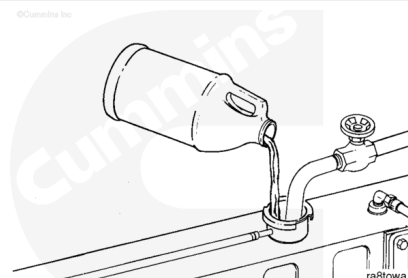


⚠ CAUTION ⚠

Fleetguard® RESTORE™ contains no antifreeze. Do not allow the cooling system to freeze during the cleaning operation.

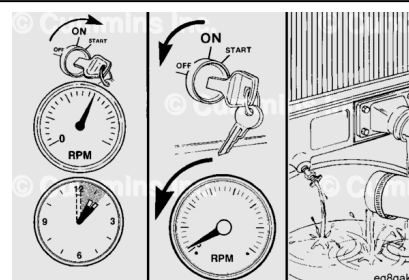
Immediately add 3.8 liters [1 gal] of Fleetguard® RESTORE™, or equivalent, for each 38 to 57 liters [10 gal] of cooling system capacity, and fill the system with plain water.

Turn the heater temperature switch to HIGH to allow maximum coolant flow through the heater core. The blower does **not** have to be in the ON position.

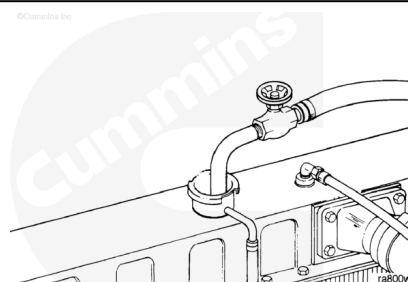


Operate the engine at a coolant temperature of at least 85°C [185°F] for 1 to 1½ hours.

Shut the engine OFF, allow to cool to 50°C [120°F], and drain the cooling system.



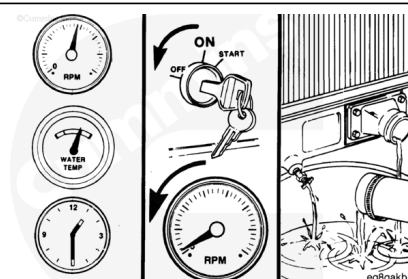
Fill the cooling system with clean water.



Operate the engine at high idle for five minutes with the coolant temperature above 85°C [185°F].

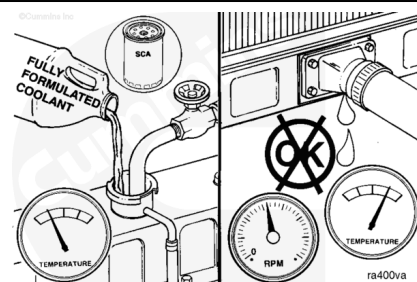
Shut the engine OFF, allow to cool to 50°C [120°F], and drain the cooling system.

If the water drained is still dirty, the system **must** be flushed again until the water is clean.

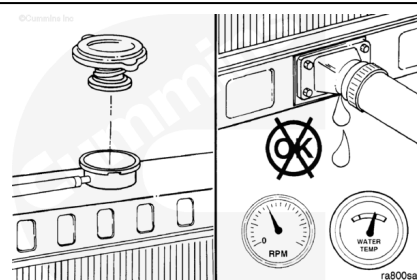


Install a new coolant filter and fill the cooling system with fully formulated coolant.

Use additional SCA to bring the coolant to the correct SCA concentration level. See the specifications in this section.



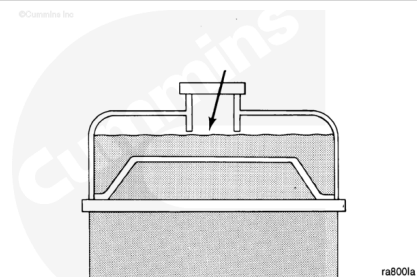
Install the radiator cap. Operate the engine until the coolant reaches a temperature of 70°C [160°F]. Check for coolant leaks.



Pressure Test

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



Check the coolant level and fill if necessary.

To aid in determining the location of the coolant leak, it could possibly be necessary to add fluorescent tracer, Part Number 3377438, to the coolant.

Add one unit of fluorescent tracer to each 38 liters [10 gal] of coolant.

Idle the engine for 5 to 10 minutes or until normal operating temperature is reached to allow the dye to circulate through the cooling system.

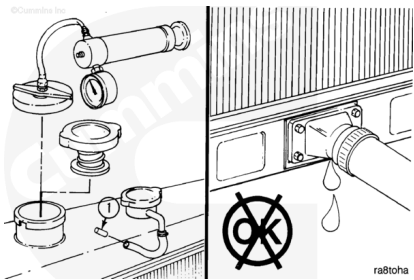
Use a high intensity black light, Part Number 3163337, or equivalent, to illuminate the dye.

If the customer experiences a coolant loss that can not be attributed to an external coolant leak or to coolant being pushed out of the overflow.

To confirm and locate external coolant leaks that occur during normal engine operation, complete this test while the engine is warm.

⚠ CAUTION ⚠

Do not apply more than 140 kPa [20 psi] air pressure to the cooling system. Excessive air pressure can damage the water pump seal.



If the radiator is equipped with a pressure relief valve, install a plug in the overflow tube (1).

Install the pressure tester on the radiator fill neck or surge tank (if equipped). Apply maximum air pressure.

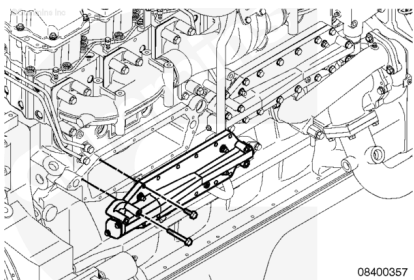
Measurements

	kpa	psi
Maximum Air Pressure	140	20

Internal

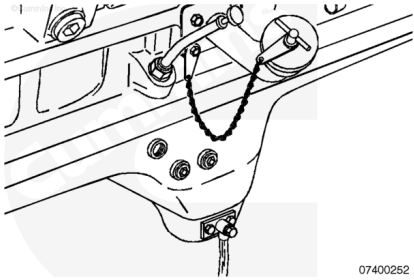
Remove the front and rear oil cooler assemblies.

Check for coolant or coolant residue in the oil passages of the oil cooler covers and the cylinder block. If coolant or residue is present, replace the oil cooler core(s). Refer to Procedure 007-007 in Section 7. (/qs3/pubsys2/xml/en/procedures/89/89-007-007.html)



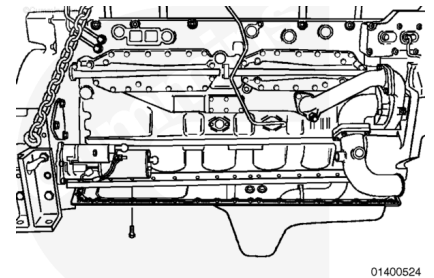
⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.



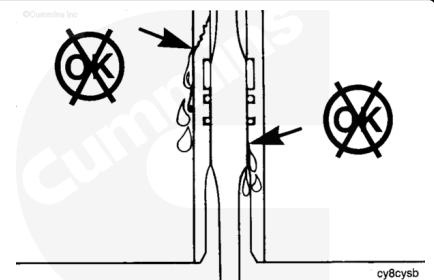
If the oil cooler cores are **not** leaking, drain the oil. Refer to Procedure 007-037 in Section 7.
(/qs3/pubsys2/xml/en/procedures/89/89-007-037.html)
Check for coolant in the oil.

If the oil contains coolant, remove the oil pan. Refer to Procedure 007-025 in Section 7.
(/qs3/pubsys2/xml/en/procedures/89/89-007-025.html)

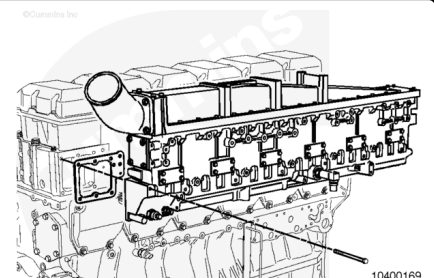


Check for leakage on the inside and outside of the cylinder liner.

Replace all cylinder liners that are leaking. Refer to Procedure 001-028 in Section 1.
(/qs3/pubsys2/xml/en/procedures/89/89-001-028.html)



Remove the intake manifold. Refer to Procedure 010-023 in Section 10. (/qs3/pubsys2/xml/en/procedures/89/89-010-023.html)

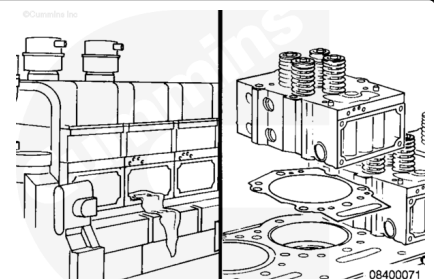


Check the intake ports and exhaust port for coolant/fluorescent dye using the black light. Presence of fluorescent dye indicates a crack.

Note : A green glow would be seen in the area where the leak has occurred. It is the coolant with tracer dye glowing from the light of the black light.

Remove the valve covers. Shine the black light in the overhead around the valve springs. If a spring deck crack is present and leaking coolant, it will glow under the black light.

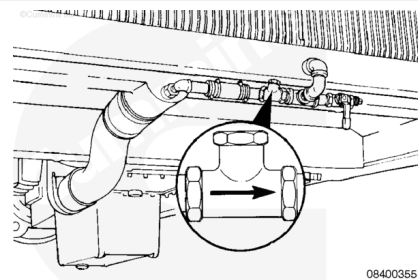
Not all cracks will easily be detected by this procedure. It is also possible to detect signs of coolant in the exhaust by looking for sign of either rust or evidence of steam cleaning in either the exhaust port in the head or its corresponding position in the exhaust manifold.



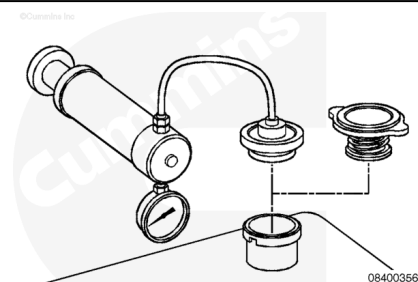
If coolant or fluorescent dye is found in the intake/exhaust ports, remove the cylinder head and test for defects. Replace if defective. Refer to Procedure 002-004 in Section 2.
(/qs3/pubsys2/xml/en/procedures/89/89-002-004.html)

External

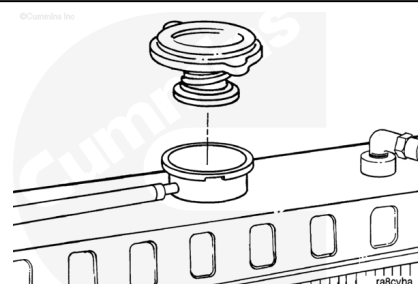
Check coolant lines, hoses, and fittings for leaks. Replace any damaged components.



Remove the pressure test equipment and the plug from the pressure relief valve overflow tube, if equipped.



Replace the radiator cap.

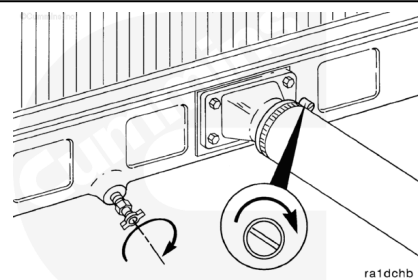


Fill

Close the radiator draincocks.

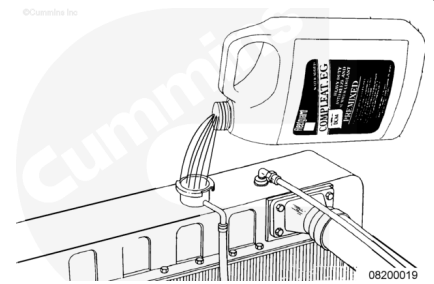
Install the lower radiator hose(s). Tighten the hose clamps.

Torque Value: 5 n·m [40 in-lb]

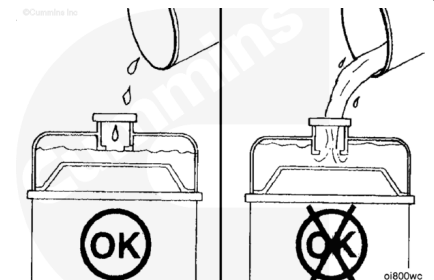


Cummins Inc. recommends using either a 50/50 mixture of good quality water and fully formulated antifreeze, or fully formulated coolant when filling the cooling system.

Use the correct units of SCA to obtain the correct cooling system protection.

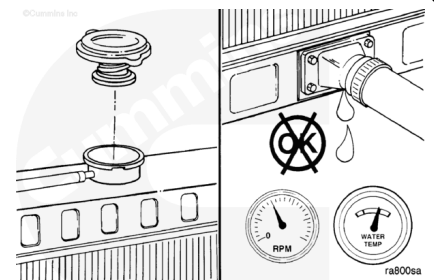


Fill the cooling system with coolant to the bottom of the fill neck in the radiator fill (or expansion) tank.



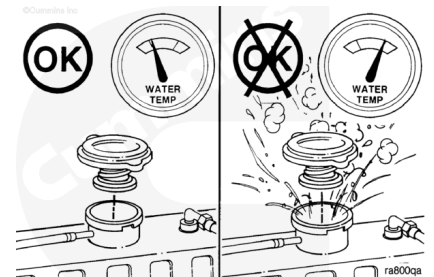
Replace the radiator cap.

Operate the engine until the coolant reaches a temperature of 70°C [160°F]. Check for leaks.



⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



Shut the engine OFF. Allow it to cool.

Check the coolant level. Fill if necessary.